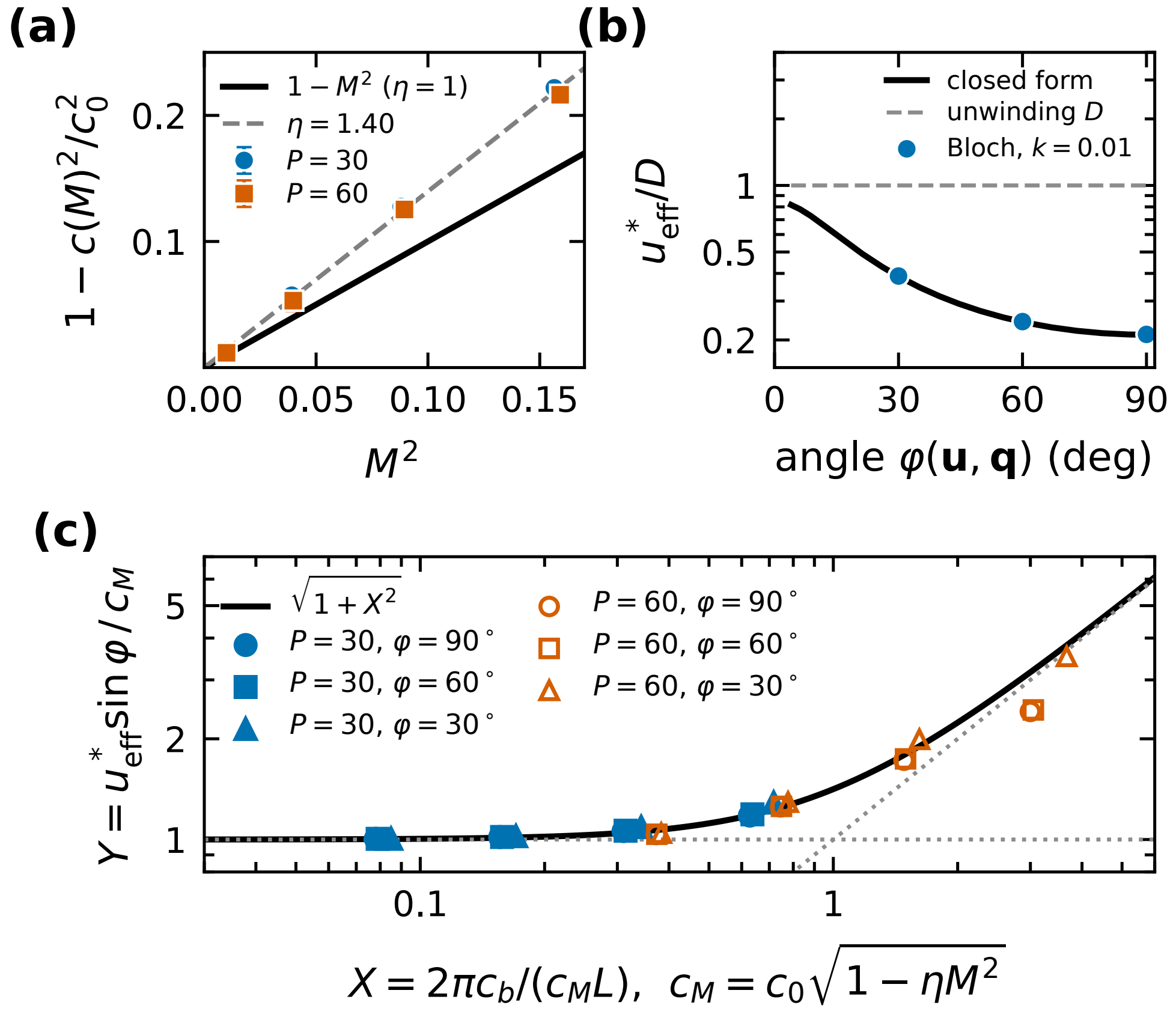
(a)
$1 - c(M)^2/c_0^2$
0.2
0.1
0.00
0.05
0.10
0.15
M^2
$1 - M^2$ ($\eta = 1$)
$\eta = 1.40$
$P = 30$
$P = 60$
(b)
u^*_{eff}/D
1
0.5
0.2
0
30
60
90
angle $\varphi(\mathbf{u}, \mathbf{q})$ (deg)
closed form
unwinding D
Bloch, $k = 0.01$
(c)
$Y = u^*_{\mathrm{eff}} \sin\varphi / c_M$
5
2
1
0.1
1
$\sqrt{1 + X^2}$
$P = 30$, $\varphi = 90^\circ$
$P = 30$, $\varphi = 60^\circ$
$P = 30$, $\varphi = 30^\circ$
$P = 60$, $\varphi = 90^\circ$
$P = 60$, $\varphi = 60^\circ$
$P = 60$, $\varphi = 30^\circ$
$X = 2\pi c_b/(c_M L), \quad c_M = c_0\sqrt{1 - \eta M^2}$